



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO NSA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

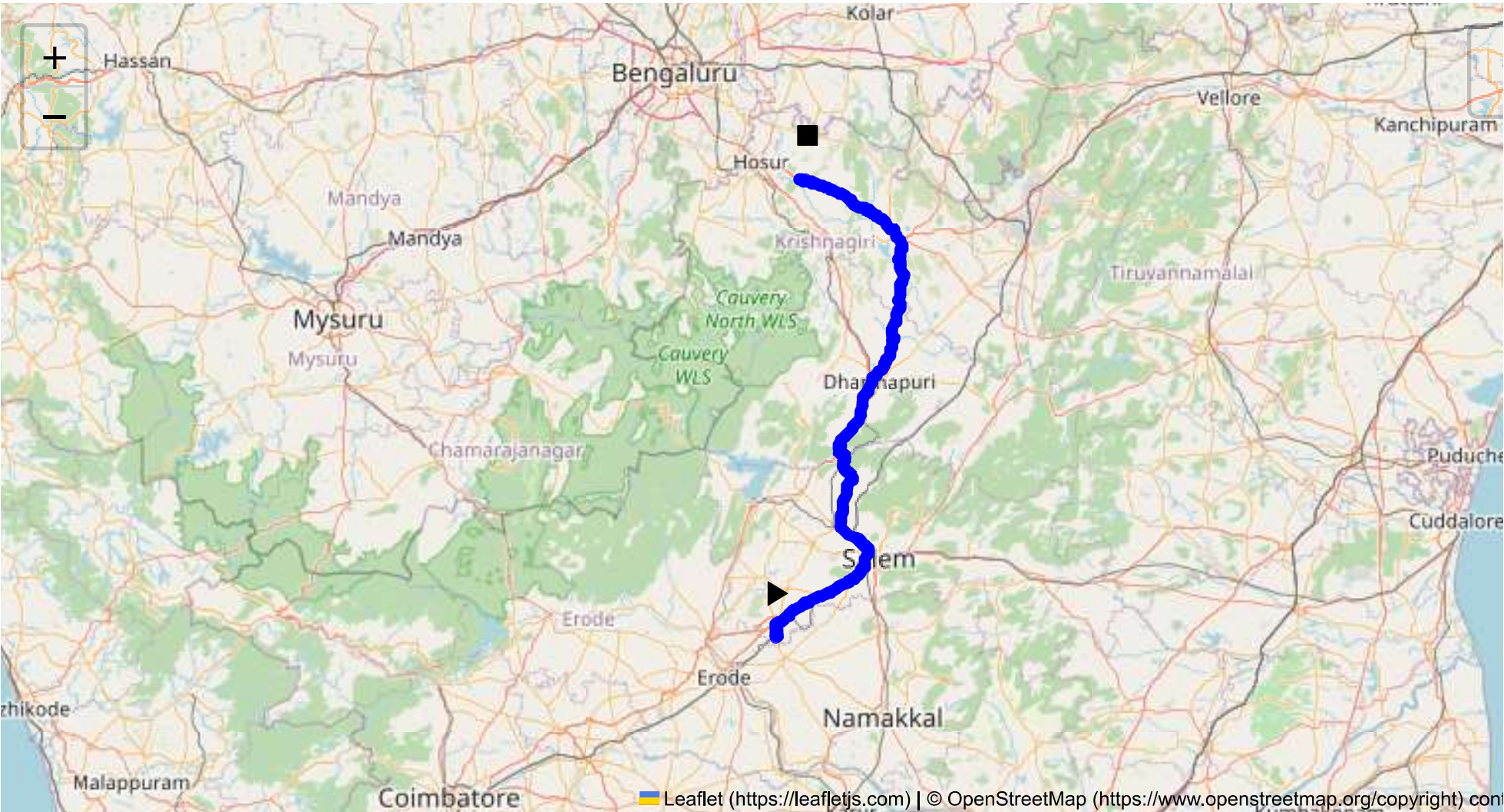
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 190.60 km
Estimated Duration: 3.4 hours
Adjusted Duration (Heavy Vehicle): 4.2 hours
Start: (11.4381, 77.8734)
End: (12.680751, 77.952324)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W Sangagiri, Tamil Nadu to NH 47 Hosur Road in Hosur spans approximately 190.60 kilometers. The typical path traverses a combination of highways and local roads, primarily following National Highway (NH) 44 and NH 544. The journey progresses through Coimbatore, Salem, and Dharmapuri districts before reaching Hosur.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** Tamil Nadu experiences a tropical climate with hot summers, a monsoon season from June to September, and mild winters.
- **Potential Hazards:** During the monsoon, heavy rains can cause flooding and landslides, especially in hilly areas near Dharmapuri. Cyclones, though less frequent, can impact coastal regions and peripheral areas.

3. Traffic Patterns

- **Peak Hours:** Generally concentrated in the mornings (8 AM to 11 AM) and evenings (5 PM to 8 PM), especially near major cities like Salem and Dharmapuri.
- **Congestion-Prone Areas:** Traffic congestion is often significant near Salem city and around urban junctions, including toll plazas and crossing points along NH 44.

4. Assessment of Road Quality and Infrastructure

- **NH Roads:** NH 44 and NH 544 are generally well-maintained, with good road quality, though maintenance work could temporarily cause disruptions.
- **Infrastructure:** Road infrastructure includes standard safety features like guardrails and signage. However, some rural road stretches may be narrower with less shoulder area.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route Suggestions:** In case of emergencies or roadblocks, exploring parallel state highways or by-lanes that link back to NH 44 can be a viable option. State Highway 79 provides another connection towards Hosur via Krishnagiri which can be considered as an alternative.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Regulations:** Truck drivers must adhere to the Motor Vehicles Act norms, obtain necessary permits for hazardous material transport, and comply with any night-time road restrictions. Safety protocols for spill management and vehicular maintenance are mandated.

7. Overview of Historical Incidents

- Previous incidents have included truck rollovers and fire hazards, primarily near Salem where congestion is significant.
- Reports indicate negligence in adherence to road signs and signals as causes for some past accidents, emphasizing the need for cautious driving.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Sections through Salem and near protected areas close to Dharmapuri involve ecological sensitivity.
- **Environmental Impact:** Adherence to low-emission protocols and prohibitions in forest reserve areas is crucial to minimize environmental harm.

9. Analysis of Communication Coverage

- **Coverage:** Mobile communication is generally reliable along NHs, with potential short dead zones in rural or forested segments near Dharmapuri.
- **Communication Needs:** Preparing for network failures by equipping trucks with satellite phones or radios is advisable.

10. Estimated Emergency Response Times

- **Near Salem and Dharmapuri:** Emergency services might arrive within 30-45 minutes due to developed infrastructure.
- **Rural Segments:** This could extend to 1-2 hours where services are sparse.

12. Overall Summary of Risk Assessment

This route presents moderate risk, primarily due to weather-related hazards during the monsoon, traffic congestion in urban areas, and potential communication interruptions. Key mitigation strategies involve attentive driving during peak hours, preparation for weather impacts, adherence to legal and environmental norms, and use of technological aids for communication.

Truck drivers should be well-informed about alternative routes, emergency contacts, and on-road assistance to manage possible risks effectively while transporting hazardous materials. Continuous updates on traffic and weather conditions can enhance overall safety during transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	Medium	11.46765, 77.87037	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47986, 77.88397	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53501, 77.95707	30 KM/Hr
11	Turn	Medium	11.53601, 77.95856	30 KM/Hr
12	Turn	Medium	11.62403, 78.10893	30 KM/Hr
13	Turn	Medium	11.62917, 78.11648	30 KM/Hr
14	Turn	High	11.64180, 78.11808	15 KM/Hr
15	Turn	High	11.64325, 78.11993	15 KM/Hr
16	Turn	Medium	11.67865, 78.12803	30 KM/Hr
17	Turn	Medium	11.68201, 78.12639	30 KM/Hr
18	Turn	Medium	11.69519, 78.10775	30 KM/Hr
19	Turn	Medium	11.73958, 78.05105	30 KM/Hr
20	Turn	Medium	11.75835, 78.05016	30 KM/Hr
21	Turn	Medium	11.77934, 78.05765	30 KM/Hr
22	Turn	Medium	11.86847, 78.08367	30 KM/Hr
23	Turn	Medium	11.87942, 78.08197	30 KM/Hr
24	Turn	Medium	11.91177, 78.05854	30 KM/Hr
25	Turn	Medium	11.92585, 78.06507	30 KM/Hr
26	Turn	Medium	11.93319, 78.06419	30 KM/Hr
27	Turn	Medium	11.93557, 78.06103	30 KM/Hr
28	Turn	Medium	11.93704, 78.05402	30 KM/Hr
29	Turn	Medium	11.95664, 78.04781	30 KM/Hr
30	Turn	High	11.95818, 78.04978	15 KM/Hr
31	Turn	Medium	11.96090, 78.04976	30 KM/Hr
32	Turn	High	11.96399, 78.05135	15 KM/Hr
33	Turn	Medium	11.96496, 78.04988	30 KM/Hr
34	Turn	High	11.96675, 78.04939	15 KM/Hr
35	Turn	Medium	11.96791, 78.05070	30 KM/Hr
36	Turn	Medium	12.16983, 78.16769	30 KM/Hr
37	Turn	Medium	12.19881, 78.18157	30 KM/Hr
38	Turn	Medium	12.22459, 78.19136	30 KM/Hr
39	Turn	Medium	12.24731, 78.19601	30 KM/Hr
40	Turn	Medium	12.25060, 78.19977	30 KM/Hr
41	Turn	Medium	12.25329, 78.20039	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
42	Turn	Medium	12.29110, 78.19909	30 KM/Hr
43	Turn	Medium	12.39904, 78.21764	30 KM/Hr
44	Turn	Medium	12.40156, 78.21845	30 KM/Hr
45	Turn	Medium	12.40589, 78.22275	30 KM/Hr
46	Turn	Medium	12.42428, 78.22671	30 KM/Hr
47	Turn	Medium	12.45154, 78.21792	30 KM/Hr
48	Turn	Medium	12.50553, 78.22318	30 KM/Hr
49	Turn	Medium	12.59749, 78.13593	30 KM/Hr
50	Turn	Medium	12.63696, 78.07061	30 KM/Hr
51	Turn	Medium	12.67143, 77.97400	30 KM/Hr
52	Blind Spot	Blind Spot	12.68367, 77.93734	10 KM/Hr
53	Turn	High	12.68387, 77.93773	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
6	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
8	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
9	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
16	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
17	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
18	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
20	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
21	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
22	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
23	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
24	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
25	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
28	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
29	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
30	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
31	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
32	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
34	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
40	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
41	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
42	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium
48	pharmacy	Mithran Medicals	12.6837657, 77.93838	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
10	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
12	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
13	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
14	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
15	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
19	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
26	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
27	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
33	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
35	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
36	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
37	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
38	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
39	school	Krishland Higher Secondary School	12.5181566, 78.2178556	30 km/h	Medium
43	school	Sri Saraswathi Vidyalaya	12.5620799, 78.183049	30 km/h	Medium
44	college	Government Medical College	12.582332, 78.1496363	30 km/h	Medium
45	school	Crescent Matriculation Higher Seconddary School	12.6003724, 78.132787	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
46	college	Jeeva College of Nursing	12.6067058, 78.1191944	30 km/h	Medium
47	school	New Little Flower Matriculation Higher Secondary School	12.6607923, 78.0169012	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47986, 77.88397



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62917, 78.11648



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64325, 78.11993



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67865, 78.12803



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68201, 78.12639



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69519, 78.10775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77934, 78.05765



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91177, 78.05854



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93319, 78.06419



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93557, 78.06103



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93704, 78.05402



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95664, 78.04781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96090, 78.04976



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96496, 78.04988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.16983, 78.16769



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22459, 78.19136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24731, 78.19601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25060, 78.19977



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25329, 78.20039



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29110, 78.19909



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39904, 78.21764



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40156, 78.21845



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40589, 78.22275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.42428, 78.22671



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.45154, 78.21792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50553, 78.22318



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59749, 78.13593



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.63696, 78.07061



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.67143, 77.97400



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.68367, 77.93734



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.68387, 77.93773